

1. be able to write down and use the definition of an indefinite integral;
2. be able to evaluate indefinite integrals.

Assigned Problems

1. Stewart, Exercises 5.4 p 415: Number 1, all the odd numbers from 7 to 21, 39, 42, 48, 53.
 2. Stewart, Exercises 5.5 p 425: All the odd numbers from 1 to 53 and from 59 to 73.
Integrate without making a substitution.
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THEME 6: VECTOR ALGEBRA

Unit 6.1 Three-dimensional coordinate systems

Source: Stewart, Section 12.1 pp 830 - 836.

Learning objectives

On completion of this unit you should

1. be able to explain what a three-dimensional rectangular coordinate system is;
2. be able to calculate the distance between two points in the three-dimensional space;
3. be able to find the projections of a given point on coordinate planes or coordinate axes;
4. be able to write down the equation of a sphere with given centre and radius.

Assigned Problems

Stewart, Exercises 12.1 p 835: Numbers 3, 4, 11, 12, 15, 19.

Unit 6.2 Vectors

Source: Stewart, Section 12.2 pp 836 - 846.

Learning objectives

On completion of this unit you should

1. know what a vector is;
2. be able to represent vectors geometrically;
3. understand that a vector has a unique direction and length but not a unique position;
4. be able to represent vectors algebraically;
5. be able to write down a vector with given initial and end points;

6. be able to find the length of a vector;
7. be able to write down and use the definitions of vector addition and scalar multiplication;
8. be able to write down and use the properties of vectors;
9. be able to write down and use the definition of parallel vectors;
10. be familiar with the standard basis vectors in two- and three-dimensional space and be able to express any vector in terms of these;
11. know what is meant by a unit vector;
12. be able to find unit vectors in the same or opposite direction of a given non-zero vector.

Assigned Problems

Stewart, Exercises 12.2 p 843: Numbers 3, 4, 7, 9, 15, 19, 21, 22, 23, 26, 29.

Unit 6.3 The dot product

Source: Stewart, Section 12.3, pp 847 - 854.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the dot product;
2. be able to determine the dot product of given vectors;
3. be able to write down and use the properties of the dot product;
4. be able to use the dot product to find the angle between two given vectors;
5. be able to use the dot product to determine whether two vectors are perpendicular (orthogonal);
6. be able to determine directional angles and cosines;
7. be able to find scalar and vector projections.

Remark: You do not have to do any proofs.

Assigned Problems

Stewart, Exercises 12.3 p 852: Numbers 1, 4, 7, 11, 12, 15, 19, 21, 23, 25, 27, 29, 33, 38, 39, 43, 45, 46.

Unit 6.4 The cross product

Source: Stewart, Section 12.4 pp 855 - 863.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the cross product of two vectors;
2. be able to find the cross product of two given vectors;
3. be able to find a vector orthogonal to two vectors;
4. be able to determine if two vectors are parallel using the cross product;
5. be able to write down and use the properties of the cross product;
6. know how to find the area of a parallelogram using the cross product.

Remarks

1. Cross products are only defined for three-dimensional vectors.
2. Note that the order of the vectors when calculating the cross product is important.
3. The introduction of a determinant is just to make it easier to calculate the cross product.
4. You do not have to know any proofs.

Assigned Problems

Stewart, Exercises 12.4 p 861: Numbers 1, 3, 7, 13, 19, 20, 27, 43.